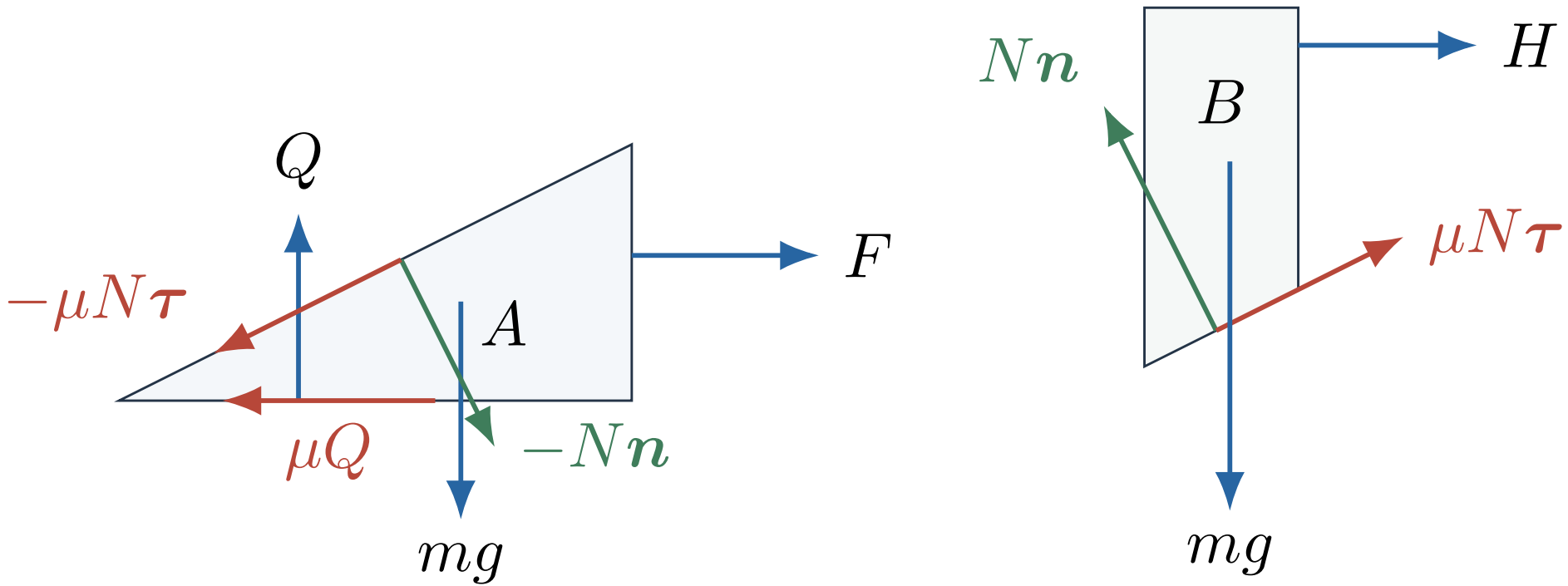

$v > 0$：接触摩擦沿相对滑动反方向